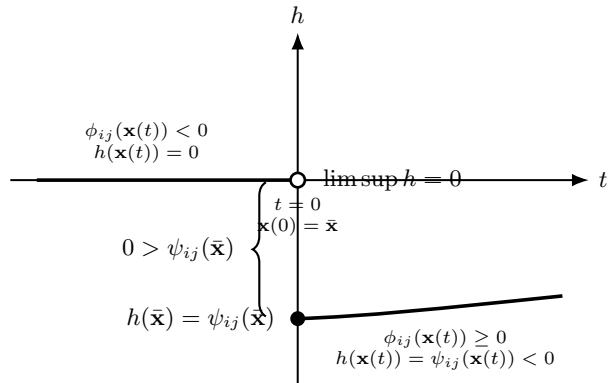


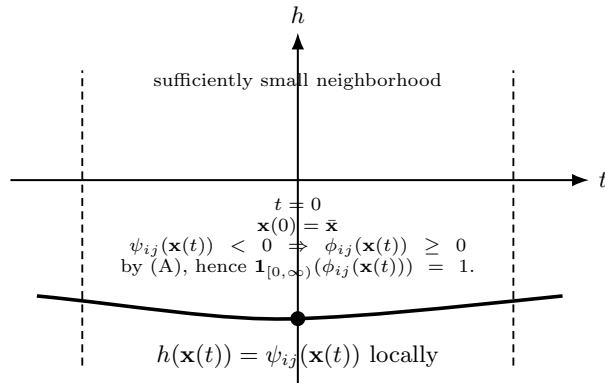
$$h(\mathbf{x}) \equiv \psi_{ij}(\mathbf{x}) \mathbf{1}_{[0, \infty)}(\phi_{ij}(\mathbf{x})), \quad \phi_{ij}(\bar{\mathbf{x}}) = 0, \quad \psi_{ij}(\bar{\mathbf{x}}) < 0.$$

(a) Without condition (A): not upper semicontinuous



A nearby inactive branch reaches 0, which is strictly above the value at $\bar{\mathbf{x}}$.

(b) Under (A): the zero-valued branch disappears locally



Near $\bar{\mathbf{x}}$, the product coincides with the continuous function ψ_{ij} .